

CHEMISTRY

## Use of Chemistry in day to day life

Use of phenolphthalein - Detection of nature (acid / base) of waste water.

\* Used in anticorruption Dept.

\* Used in soil testing

Use of  $H_2SO_4$  - Used in manufacturing of ammonium sulphate (fertilizer)

\* It is used in refining petroleum.

\* It is used in production of various salt.

\* It is very important acid for chemical detection (testing).

\* It is very widely used in synthesis of detergents.

MATHS



26/11/25

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Q) A room is 7m long and 5m broad. It has one door measuring (2m by 1.5m) and two windows, each measuring (1.5m by 1m). The cost of painting the walls of the room at ₹80/m<sup>2</sup> is ₹5280. Find the height of the room.

Length of the room = 7m  
Breadth = 5m

Total cost of painting = ₹5280

Rate of painting = ₹80/m<sup>2</sup>

Area of the painted part of 4 walls

$$= \left( \frac{\text{total cost}}{\text{rate/m}^2} \right) \text{m}^2 = \left( \frac{5280}{80} \right) \text{m}^2$$

$$= 66 \text{ m}^2$$

$$\text{Area of 1 door} = \left( 2 \times \frac{3}{2} \right) \text{m}^2 = 3 \text{ m}^2$$

$$\begin{aligned} \text{Area of 2 windows} &= \left\{ 2 \times \left( \frac{3}{2} \times 1 \right) \right\} \text{m}^2 \\ &= \left\{ 2 \times \frac{3}{2} \right\} \text{m}^2 = 3 \text{ m}^2 \end{aligned}$$



Total area of 1 door and 2 windows  
 $= (3+3) \text{ m}^2 = 6 \text{ m}^2$

Total area of 4 walls  $(66+6) \text{ m}^2 = 72 \text{ m}^2$

But, total area of 4 walls  $= 2(l+b) \times h$   
 $= 2(7+5) \times h$   
 $= (24h) \text{ m}^2$

$$\therefore 24h = 72 \Rightarrow h = \frac{72}{24} = 3 \text{ m}$$

Hence height of the room is 3 m

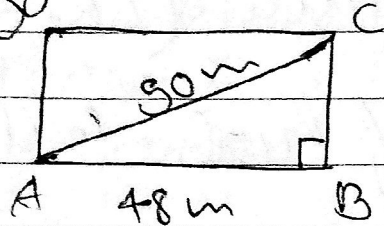
EX-20A

1) Find the area of the rect. whose dimensions are:

(i)  $l = 24.5 \text{ m}$ ,  $b = 18 \text{ m}$

$$\begin{aligned} \text{Area of rect.} &= (l \times b) \\ &= \left( \begin{array}{r} 24.5 \\ \times 18 \\ \hline 196 \\ + 1960 \\ \hline 441 \end{array} \right) \text{ m}^2 = 441 \text{ m}^2 \end{aligned}$$

- 2) Find the area of a rect. plot one side of which is 48m and its diagonal is 50m.



In right  $\triangle ABC$ ;

$$AC^2 = AB^2 + BC^2 \text{ [Pythagoras the.]}$$

$$\Rightarrow 50^2 = (48)^2 + BC^2$$

$$\Rightarrow BC^2 = (50)^2 - (48)^2$$

$$\Rightarrow BC^2 = (50+48)(50-48)$$

$$\Rightarrow BC^2 = 98 \times 2$$

$$\Rightarrow BC^2 = 196$$

$$\Rightarrow BC = \sqrt{196}$$

$$\therefore BC = 14$$

∴

Breadth of the rect. plot:  $\boxed{14}$  m

$$\therefore \text{Area of the rect. plot} = l \times b \\ = (48 \times 14) \text{m}^2 \\ = 672 \text{m}^2$$

1) ii)  $l = 12.5 \text{m}, b = 8 \text{dm}$

$$\text{Area} = l \times b \\ = (12.5 \times 8) \text{m}^2$$

$$= \left( \begin{array}{r} 255 \\ 125 \\ 10 \\ 2 \end{array} \times \begin{array}{r} 42 \\ 8 \\ 10 \\ 5 \end{array} \right) \text{m}^2 = 10 \text{m}^2$$

3) The sides of rect. field are in the ratio 4:3. If its area is  $1728 \text{m}^2$ . find the cost of fencing it at ₹30/m

Let the length of the field be  $4u \text{m}$   
& breadth be  $3u \text{m}$

$$\therefore \text{area} = (4u \times 3u) \text{m}^2 = (12u^2) \text{m}^2$$

But, area =  $1728 \text{m}^2$

$$\therefore 12u^2 = 1728 \Rightarrow u^2 = \frac{1728}{12} = 144$$

$$\Rightarrow u = \sqrt{144} = 12$$



- 5) A verandah is 40 m long and 15 m broad. It is to be paved with stones each measuring  $(6 \times 8)$  dm. Find the no. of stones req.

$$\text{Area of verandah} = (40 \times 15) \text{ m}^2$$

$$= 600 \text{ m}^2$$

$$\text{Area of 1 stone} = 6 \text{ dm} \times 8 \text{ dm}$$

$$= \left( \frac{6}{10} \times \frac{8}{10} \right) \text{ m}^2 = 0.3 \text{ m}^2$$

$$\therefore \text{req. no. of stones} = \frac{\text{area of verandah}}{\text{area of stone}}$$

$$= \frac{600}{0.3} = \frac{600 \times 10}{3}$$

$$= \boxed{2000}$$

- 8) The length and breadth of a rect. piece of land are in the ratio 5:3. If the total cost of fencing it at ₹ 24/m is ₹ 9600. Find its

length and breadth.

It is